Research Notes on Mixture model

Mixture model

>> Section 1

House prices Assume that we observe the prices of N different houses. Different types of houses in different neighborhoods will have vastly different prices, but the price of a particular type of house in a particular neighborhood (e.g., three-bedroom house in moderately upscale neighborhood) will tend to cluster fairly closely around the mean. One possible model of such prices would be to assume that the prices are accurately described by a mixture model with K different components, each distributed as a normal distribution with unknown mean and variance, with each component specifying a particular combination of house type/neighborhood. Fitting this model to observed prices, e.g., using the expectation-maximization algorithm, would tend to cluster the prices according to house type/neighborhood and reveal the spread of prices in each type/neighborhood. (Note that for values such as prices or incomes that are guaranteed to be positive and which tend to grow exponentially, a log-normal distribution might actually be a better model than a normal distribution.)

>> Section 2

K , N = as above φ i = 1 … K , φ = as above z i = 1 … N , x i = 1 … N = as above θ i = 1 … K = { μ i = 1 … K , σ i = 1 … K 2 } μ i = 1 … K = mean of component i σ i = 1 … K 2 = variance of component i z i = 1 … N ~ Categorical ⁡ (φ) x i = 1 … N ~ N (μ z i , σ z i 2) {\displaystyle {\begin{array}{lcl}K,N&=&{\text{as above}}\\\phi _{i=1\dots K},{\boldsymbol {\phi }}&=&{\text{as above}}\\z_{i=1\dots N},x_{i=1\dots N}&=&{\text{as above}}\\\theta _{i=1\dots K}&=&\{\mu _{i=1\dots K},\sigma _{i=1\dots K}^{2}\}\\\mu _{i=1\dots K}&=&{\text{mean of component }}i\\\sigma _{i=1\dots K}^{2}&=&{\text{variance of component }}i\\z_{i=1\dots N}&\sim &\operatorname {Categorical} ({\boldsymbol {\phi }})\\x_{i=1\dots N}&\sim &{\mathcal {N}}(\mu _{z_{i}},\sigma _{z_{i}}^{2})\end{array}}}

>> Section 3

K = { p (·) : p (·) = ∑ i = 1 n a i f i (· ; θ i) , a i > 0 , ∑ i = 1 n a i = 1 , f i (· ; θ i) ∈ J ∀ i , n } {\displaystyle K=\left\{p(\cdot):p(\cdot)=\sum _{i=1}^{n}a_{i}f_{i}(\cdot ;\theta _{i}),a_{i}>0,\sum _{i=1}^{n}a_{i}=1,f_{i}(\cdot ;\theta _{i})\in J\ \forall i,n\right\}}

>> Section 4

Financial returns often behave differently in normal situations and during crisis times. A mixture model for return data seems reasonable. Sometimes the model used is a jump-diffusion model, or as a mixture of two normal distributions. See Financial economics § Challenges and criticism and Financial risk management § Banking for further context.

>> Section 5

Topics in a document Assume that a document is composed of N different words from a total vocabulary of size V, where each word corresponds to one of K possible topics. The distribution of such words could be modelled as a mixture of K different V-dimensional categorical distributions. A model of this sort is commonly termed a topic model. Note that expectation maximization applied to such a model will typically fail to produce realistic results, due (among other things) to the excessive number of parameters. Some sorts of additional assumptions are typically necessary to get good results. Typically two sorts of additional components are added to the model:

>> Section 6

K = number of mixture components N = number of observations θ i = 1 … K = parameter of distribution of observation associated with component i φ i = 1 … K = mixture weight, i.e., prior probability of a particular component i φ = K -dimensional vector composed of all the individual φ 1 … K ; must sum to 1 z i = 1 … N = component of observation i x i = 1 … N = observation i F (x | θ) = probability distribution of an observation, parametrized on θ z i = 1 … N ~ Categorical ⁡ (φ) x i = 1 … N | z i = 1 … N ~ F (θ z i) {\displaystyle {\begin{array}{lcl}K&=&{\text{number of mixture components}}\\N&=&{\text{number of observations}}\\\theta _{i=1\dots K}&=&{\text{parameter of distribution of observation associated with component }}i\\\phi _{i=1\dots K}&=&{\text{mixture weight, i.e., prior probability of a particular component }}i\\{\boldsymbol {\phi }}&=&K{\text{-dimensional vector composed of all the individual }}\phi _{1\dots K}{\text{; must sum to 1}}\\z_{i=1\dots N}&=&{\text{component of observation }}i\\x_{i=1\dots N}&=&{\text{observation }}i\\F(x|\theta)&=&{\text{probability distribution of an observation, parametrized on }}\theta \\z_{i=1\dots N}&\sim &\operatorname {Categorical} ({\boldsymbol {\phi }})\\x_{i=1\dots N}|z_{i=1\dots N}&\sim &F(\theta _{z_{i}})\end{array}}}

>> Section 7

p (θ) = ∑ i = 1 K φ i N (μ i , Σ i) {\displaystyle p({\boldsymbol {\theta }})=\sum _{i=1}^{K}\phi _{i}{\mathcal {N}}({\boldsymbol {\mu }}_{i},{\boldsymbol {\Sigma }}_{i})}

>> Section 8

Parameter estimation and system identification Parametric mixture models are often used when we know the distribution Y and we can sample from X, but we would like to determine the ai and θi values. Such situations can arise

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in studies in which we sample from a population that is composed of several distinct subpopulations. It is common to think of probability mixture modeling as a missing data problem. One way to understand this is to assume that the data points under consideration have "membership" in one of the distributions we are using to model the data. When we start, this membership is unknown, or missing. The job of estimation is to devise appropriate parameters for the model functions we choose, with the connection to the data points being represented as their membership in the individual model distributions. A variety of approaches to the problem of mixture decomposition have been proposed, many of which focus on maximum likelihood methods such as expectation maximization (EM) or maximum a posteriori estimation (MAP). Generally these methods consider separately the questions of system identification and parameter estimation; methods to determine the number and functional form of components within a mixture are distinguished from methods to estimate the corresponding parameter values. Some notable departures are the graphical methods as outlined in Tarter and Lock and more recently minimum message length (MML) techniques such as Figueiredo and Jain and to some extent the moment matching pattern analysis routines suggested by McWilliam and Loh (2009).